GLOBAL ACADEMY OF TECHNOLOGY
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LATEX
BCS24457D
LAB MANUAL

IV SEMESTER

2025 - 2026

Prepared by

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Dept. of CSE

Approved by

Dr. R Rajkumar
Professor & Head Dept. of CSE

B.E - COMPUTER SCIENCE AND ENGINEERING

SEMESTER - IV

COURSE: Technical Writing Using LATEX

Course Code:	**BCS24457D**	**CIE Marks:**	**50**
Hours/Week (L: T: P):	**0:0:2**	**SEE Marks:**	**50**
Total Hours:	**28**	**Examination Hours:**	**3 hours**
No. of Credits: 01			

Course Learning Objectives:

The course will enable students to:

CLO1	To introduce the basic syntax and semantics of the LaTeX scripting language
CLO2	To understand the presentation of tables and figures in the document
CLO3	To illustrate the LaTeX syntax to represent the theorems and mathematical equations
CLO4	To make use of the libraries (Tikz, algorithm) to design the diagram and algorithms in the document

Q. No.	Question
1	Develop a LaTeX script to create a simple document that consists of 2 sections [Section1, Section2], and a paragraph with dummy text in each section. And also include header [title of document] and footer [institute name, page number] in the document.
2	Develop a LaTeX script to create a document that displays the sample Abstract/Summary.
3	Develop a LaTeX script to create a simple title page of the VTU project Report [Use suitable Logos and text formatting]
4	Develop a LaTeX script to create the Certificate Page of the Report [Use suitable commands to leave the blank spaces for user entry].
5	Develop a LaTeX script to create a document that contains the following table with proper labels (table below)
6	Develop a LaTeX script to include the side-by-side graphics/pictures/figures in the document by using the subgraph concept.
7	Develop a LaTeX script to create a document that consists of the following two mathematical equations

Sl. No	USN	Student Name	Marks		
			Subject1	Subject2	Subject3
1	4XX22XX001	Name 1	89	60	90
2	4XX22XX002	Name 2	78	45	98
3	4XX22XX003	Name 3	67	55	59

	$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{-2 \pm \sqrt{2^2-4\cdot(1)\cdot(-8)}}{2\cdot 1} = \frac{-2 \pm \sqrt{4+32}}{2}$ $\varphi_\sigma^\lambda A_t = \sum_{\pi \in C_t} \mathrm{sgn}(\pi)\varphi_\sigma^\lambda \varphi_\pi^\lambda = \sum_{\tau \in C_{\sigma t}} \mathrm{sgn}(\sigma^{-1}\tau\sigma)\varphi_\sigma^\lambda \varphi_{\sigma^{-1}\tau\sigma}^\lambda = A_{\sigma t}\varphi_\sigma^\lambda$
8	Develop a LaTeX script to demonstrate the presentation of Numbered theorems, definitions, corollaries, and lemmas in the document.
9	Develop a LaTeX script to create a document that consists of two paragraphs with a minimum of 10 citations in it and display the reference in the section
10	Develop a LaTeX script to design a simple tree diagram or hierarchical structure in the document with appropriate labels using the Tikz library.
11	Develop a LaTeX script to present an algorithm in the document using algorithm/algorithmic/algorithm2e library.
12	Develop a LaTeX script to create a simple report and article by using suitable commands and formats of user choice.

Course Outcomes: Upon successful completion of this course, students will be able to

CO1	Apply basic LaTeX command to develop simple document
CO2	Develop LaTeX script to present the tables and figures in the document
CO3	Illustrate LaTeX script to present theorems and mathematical equations in the document
CO4	Develop programs to generate the complete report with citations and a bibliography
CO5	Illustrate the use of Tikz and algorithm libraries to design graphics and algorithms in the document

Mapping of CO-PO:

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	-	-	-	3	-	-	-	-	-	-	-	-
CO2	2	2	3	-	3	-	-	-	-	-	1	2	-
CO3	3	-	-	-	3	-	-	-	-	-	-	-	-
CO4	3	2	3	-	3	-	-	-	-	-	1	2	-
CO5	3	-	-	-	3	-	-	-	-	-	-	-	-

Low-1: Medium-2: High-3

LAB EVALUATION PROCESS

Continuous Internal Evaluation (CIE1)		
S. No	**Activity**	**Marks**
1	Average of Weekly Entries	20
2	Lab Internal Assessment	30
TOTAL		**50**

Continuous Internal Evaluation (CIE2)		
S. No	**Activity**	**Marks**
1	Average of Weekly Entries	20
2	Lab Internal Assessment	30
TOTAL		**50**

Average of CIE1 and CIE2	**50**

Semester End Examination (SEE) Evaluation		
S. No	**Activity**	**Marks**
1	Writeup + Execution + Viva Voce	100
TOTAL		**50**

ASSESSMENT

a) Continuous Internal Evaluation (CIE):

	Q. No.	Writeup	Execution	Viva	Total Marks	Reduced to 40
CIE1	**Q1 to Q6**	**15**	**70**	**15**	**100**	**30**
CIE2	**Q7 to Q12**	**15**	**70**	**15**	**100**	**30**
Record						**20**
Total						**50**
		Final CIE Marks = CIE1 + CIE2 =30 M + 20 M= 50M				

b) Semester End Examination (SEE) Evaluation:

Question No.	Writeup	Execution	Viva	Total Marks
1	15	70	15	100
Total Marks	**15**	**70**	**15**	**Reduce to 50**

Lab Rubrics

Rubrics for Evaluation of Record

Attribute	Max Marks	Good	Satisfactory	Poor
Writeup	8	7–8: All programs written correctly without syntax errors.	4–6: Minor syntax/logical errors.	0–3: Major syntax errors or incomplete work.
Execution of Program	8	7–8: Programs execute correctly with expected output.	4–6: Executes with minor issues or incomplete output.	0–3: Programs fail to execute properly.
Viva Voce	4	4: Clear explanation of concepts and commands.	2–3: Partial understanding.	0–1: Unable to explain concepts.

Rubrics for Evaluation of Internal Test

Attribute	Max Marks	Good	Satisfactory	Poor
Writeup	15	13–15: Correct syntax and proper structure.	7–12: Minor errors in syntax or formatting.	0–6: Major syntax errors and poor structure.
Execution of Program	70	60–70: Programs execute correctly and produce expected output.	35–59: Executes with some errors or incomplete output.	0–34: Programs fail to execute or incorrect output.
Viva Voce	15	13–15: Clear understanding of concepts.	7–12: Partial understanding.	0–6: Unable to explain concepts.

LAB PROGRAMS

1. Develop a LaTeX script to create a simple document that consists of 2 sections [Section1, Section2], and a paragraph with dummy text in each section. And also include header [title of document] and footer [institute name, page number] in the document

Solution:

```
\documentclass{article}
\usepackage{fancyhdr}% Header and Footer
\pagestyle{fancy}
\fancyhf{} % Clear default header and footer
\fancyhead[c]{My First Program}
\fancyfoot[L]{Global Academy of Technology} % Footer - Institute name and page number
\fancyfoot[R]{\thepage}
\begin{document}
\thispagestyle{fancy}
% Section 1
\begin{center}
\section*{Introduction}
\end{center}
\section{Computer}
A computer is an electronic machine that accepts data as input, processes it according to a set of instructions, and produces meaningful information as output. It works based on programs that tell it what tasks to perform. Computers are capable of performing calculations, storing large amounts of data, and executing complex operations at very high speed and accuracy. They are widely used in fields such as education, business, healthcare, entertainment, and communication. Because of their speed, reliability, and versatility, computers have become an essential part of modern life.
% Section 2
\section{Parts of the computer}
The main parts of a computer are divided into hardware components that work together to perform tasks. The central processing unit (CPU) is the brain of the computer, responsible for processing instructions and controlling operations. The memory unit stores data and instructions temporarily (RAM) or permanently (storage devices like hard disks or SSDs). Input devices such as the keyboard and mouse are used to enter data, while output devices like the monitor and printer display or produce results. All these parts work together to ensure that the computer functions efficiently and accurately.
\end{document}
```

Output:

My First Program

Introduction

1 Computer

A computer is an electronic machine that accepts data as input, processes it according to a set of instructions, and produces meaningful information as output. It works based on programs that tell it what tasks to perform. Computers are capable of performing calculations, storing large amounts of data, and executing complex operations at very high speed and accuracy. They are widely used in fields such as education, business, healthcare, entertainment, and communication. Because of their speed, reliability, and versatility, computers have become an essential part of modern life.

2 Parts of the computer

The main parts of a computer are divided into hardware components that work together to perform tasks. The central processing unit (CPU) is the brain of the computer, responsible for processing instructions and controlling operations. The memory unit stores data and instructions temporarily (RAM) or permanently (storage devices like hard disks or SSDs). Input devices such as the keyboard and mouse are used to enter data, while output devices like the monitor and printer display or produce results. All these parts work together to ensure that the computer functions efficiently and accurately.

Global Academy of Technology 1

2. Develop a LaTeX script to create a document that displays the sample Abstract/Summary.

Solution:

```
\documentclass{article}
\begin{document}
% Title and Author
\title{Sample Abstract}
\author{Thejeswini A M}
\date{\today} % Remove date
\maketitle
% Abstract Section
\section*{Abstract}
Computer Science is the study of computers and computational systems. It focuses on the design, development, and analysis of software and hardware used to process information. The field includes topics such as programming, data structures, algorithms, artificial intelligence, computer networks, databases, and cybersecurity. Computer Science plays an important role in modern society by enabling innovations in communication, healthcare, education, business, and entertainment. With rapid technological advancements, it continues to evolve and provide solutions to complex real-world problems. This discipline not only improves efficiency and automation but also contributes to scientific research and digital transformation across various industries.
\end{document}
```

Output:

Sample Abstract

Thejeswini A M

February 16, 2026

Abstract

Computer Science is the study of computers and computational systems. It focuses on the design, development, and analysis of software and hardware used to process information. The field includes topics such as programming, data structures, algorithms, artificial intelligence, computer networks, databases, and cybersecurity. Computer Science plays an important role in modern society by enabling innovations in communication, healthcare, education, business, and entertainment. With rapid technological advancements, it continues to evolve and provide solutions to complex real-world problems. This discipline not only improves efficiency and automation but also contributes to scientific research and digital transformation across various industries.

1

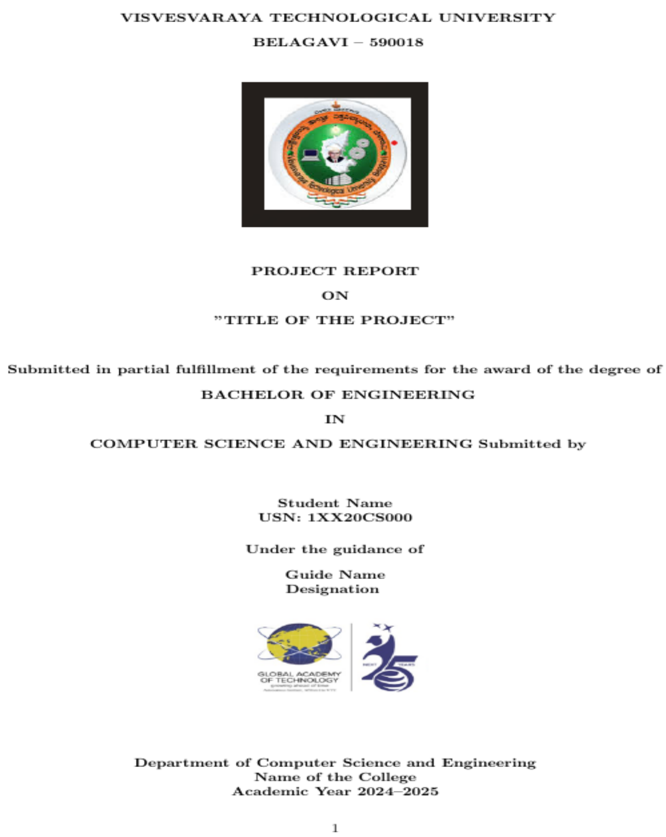
3. Develop a LaTeX script to create a simple title page of the VTU project Report [Use suitable Logos and text formatting]

Solution:

```
\documentclass{report}
\usepackage{graphicx}
\usepackage{geometry}
\geometry{margin=1in}
\begin{document}
\centering
\vspace{1cm}
\textbf{ VISVESVARAYA TECHNOLOGICAL UNIVERSITY}\\
\vspace{0.3cm}
\textbf{ BELAGAVI – 590018}\\
\vspace{1cm}
% VTU Logo
\centering
\includegraphics[width=0.25\textwidth]{vtu_logo.png}\\
\vspace{1cm}
\textbf{PROJECT REPORT}\\
\vspace{0.3cm}
\textbf{ON}\\
\vspace{0.3cm}
\textbf{\textbf{"TITLE OF THE PROJECT"}}\\
\vspace{1cm}
\textbf{Submitted in partial fulfillment of the requirements for the award of the degree of}\\
\vspace{0.3cm}
\textbf{ BACHELOR OF ENGINEERING}\\
\vspace{0.3cm}
\textbf{IN}\\
\vspace{0.3cm}
\textbf{ COMPUTER SCIENCE AND ENGINEERING}
\vspace{1cm}
\textbf{Submitted by}\\
\vspace{0.3cm}
\centering
```

```
    \textbf{Student Name} \\
    \textbf{USN: 1XX20CS000}
    \vspace{0.5cm}
    \textbf{Under the guidance of}\\
    \vspace{0.3cm}
    \textbf{Guide Name}\\
    \textbf{Designation}
    \vspace{0.5cm}
    % VTU Logo
    \includegraphics[width=0.25\textwidth]{clg_logo.png}
    \vfill
    \textbf{Department of Computer Science and Engineering}\\
    \textbf{Name of the College}\\
    \textbf{Academic Year 2024–2025}
\end{document}
```

Output:


VISVESVARAYA TECHNOLOGICAL UNIVERSITY

BELAGAVI – 590018

PROJECT REPORT

ON

"TITLE OF THE PROJECT"

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING Submitted by

Student Name
USN: 1XX20CS000

Under the guidance of

Guide Name
Designation

Department of Computer Science and Engineering
Name of the College
Academic Year 2024–2025

1

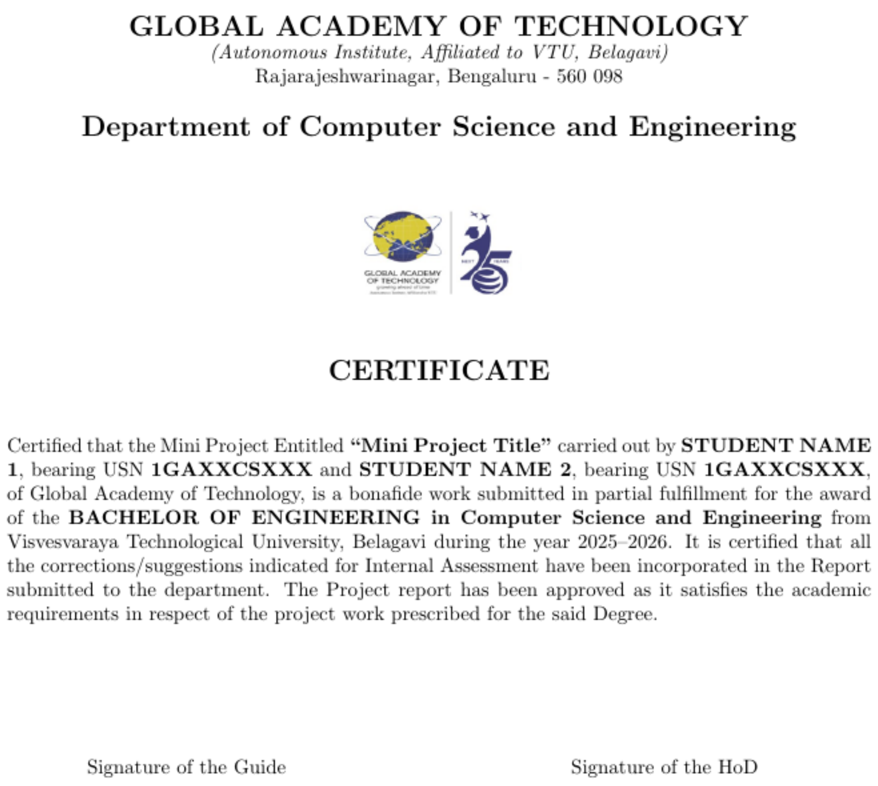
4. Develop a LaTeX script to create the Certificate Page of the Report [Use suitable commands to leave the blank spaces for user entry].

Solution:

```
\documentclass{article}
\usepackage{graphicx}
%\usepackage{setspace}
\usepackage{geometry}
\usepackage{ragged2e}
\geometry{left=1.25in,right=1.25in,top=1in,bottom=1in}
\pagestyle{empty}
\begin{document}
\begin{center}
\textbf{\Large GLOBAL ACADEMY OF TECHNOLOGY}\\
\textit{(Autonomous Institute, Affiliated to VTU, Belagavi)}\\
Rajarajeshwarinagar, Bengaluru - 560 098\\[0.5cm]
\textbf{\Large Department of Computer Science and Engineering}\\
% ---- VTU LOGO INSERTED HERE ----
\centering
\vspace{1cm}
\includegraphics[width=3cm]{clg_logo.png}
\vspace{1cm}\\
\textbf{\Large CERTIFICATE}
\end{center}
\vspace{0.5cm}
\justifying
\noindent Certified that the Mini Project Entitled \textbf{``Mini Project Title"}
carried out by \textbf{STUDENT NAME 1}, bearing USN
\textbf{1GAXXCSXXX} and \textbf{STUDENT NAME 2}, bearing USN
\textbf{1GAXXCSXXX}, of Global Academy of Technology,
is a bonafide work submitted in partial fulfillment for the award of the
\textbf{BACHELOR OF ENGINEERING in Computer Science and Engineering}
from Visvesvaraya Technological University, Belagavi during the year
2025--2026. It is certified that all the corrections/suggestions indicated
for Internal Assessment have been incorporated in the Report submitted
to the department. The Project report has been approved as it satisfies
the academic requirements in respect of the project work prescribed for
```

```
the said Degree.
\vspace{2cm}
\begin{center}
\begin{tabular}{p{6cm} p{6cm}}
Signature of the Guide & { \hspace {2cm} Signature of the HoD} \\
\end{tabular}
\end{center}
\end{document}
```

Output:



GLOBAL ACADEMY OF TECHNOLOGY

(Autonomous Institute, Affiliated to VTU, Belagavi)

Rajarajeshwarinagar, Bengaluru - 560 098

Department of Computer Science and Engineering

CERTIFICATE

Certified that the Mini Project Entitled **"Mini Project Title"** carried out by **STUDENT NAME 1**, bearing USN **1GAXXCSXXX** and **STUDENT NAME 2**, bearing USN **1GAXXCSXXX**, of Global Academy of Technology, is a bonafide work submitted in partial fulfillment for the award of the **BACHELOR OF ENGINEERING in Computer Science and Engineering** from Visvesvaraya Technological University, Belagavi during the year 2025–2026. It is certified that all the corrections/suggestions indicated for Internal Assessment have been incorporated in the Report submitted to the department. The Project report has been approved as it satisfies the academic requirements in respect of the project work prescribed for the said Degree.

Signature of the Guide   Signature of the HoD

5. Develop a LaTeX script to create a document that contains the following table with proper labels

Sl. No	USN	Student Name	Marks		
			Subject1	**Subject2**	**Subject3**
1	4XX22XX001	Name 1	89	60	90
2	4XX22XX002	Name 2	78	45	98
3	4XX22XX003	Name 3	67	55	59

Solution:

```latex
\documentclass{article}
\usepackage[a4paper,margin=1in]{geometry}
\usepackage{multirow}
\usepackage{array}
\begin{document}
\begin{center}
\textbf{\Large Student Marks Table}
\end{center}
\vspace{0.5cm}
\begin{table}[h!]
\centering
\begin{tabular}{|c|c|c|c|c|c|}
\hline
\multirow{2}{*}{\textbf{Sl. No}} &
\multirow{2}{*}{\textbf{USN}} &
\multirow{2}{*}{\textbf{Student Name}} &
\multicolumn{3}{c|}{\textbf{Marks}} \\ \cline{4-6}
& & & \textbf{Subject1} & \textbf{Subject2} & \textbf{Subject3} \\
\hline
1 & 4XX22XX001 & Name 1 & 89 & 60 & 90 \\
\hline
2 & 4XX22XX002 & Name 2 & 78 & 45 & 98 \\
\hline
3 & 4XX22XX003 & Name 3 & 67 & 55 & 59 \\
\hline
\end{tabular}
\caption{Student Marks Details}
\end{table}
\end{document}
```

Output:

Student Marks Table

Sl. No	USN	Student Name	Marks		
			Subject1	Subject2	Subject3
1	4XX22XX001	Name 1	89	60	90
2	4XX22XX002	Name 2	78	45	98
3	4XX22XX003	Name 3	67	55	59

Table 1: Student Marks Details

6. Develop a LaTeX script to include the side-by-side graphics/pictures/figures in the document by using the subgraph concept.

Solution:

```
\documentclass{article}
\usepackage[a4paper,margin=1in]{geometry}
\usepackage{graphicx}
\usepackage{subcaption} % For subfigure environment
\begin{document}
\title{Side-by-Side Figures Example}
\author{}
\date{}
\maketitle
\begin{figure}[h]
\centering
% First Subfigure
\begin{subfigure}{0.45\textwidth}
\centering
\includegraphics[width=\linewidth]{clg_logo.png}
\caption{First Image}
\label{fig:first}
\end{subfigure}
\hfill
% Second Subfigure
\begin{subfigure}{0.45\textwidth}
\centering
\includegraphics[width=\linewidth]{clg_logo.png}
\caption{Second Image}
\label{fig:second}
\end{subfigure}
\caption{Side-by-Side Graphics Using Subfigure Concept}
\label{fig:combined}
\end{figure}
\end{document}
```

Output:

Side-by-Side Figures Example

(a) First Image (b) Second Image

Figure 1: Side-by-Side Graphics Using Subfigure Concept

7. Develop a LaTeX script to create a document that consists of the following two mathematical equations

$$
\begin{aligned}
x &= \frac{-b \pm \sqrt{b^2-4ac}}{2a} \\
&= \frac{-2 \pm \sqrt{2^2-4*(1)*(-8)}}{2*1} \\
&= \frac{-2 \pm \sqrt{4+32}}{2}
\end{aligned}
\qquad
\begin{aligned}
\varphi_\sigma^\lambda A_t &= \sum_{\pi \in C_t} \operatorname{sgn}(\pi) \varphi_\sigma^\lambda \varphi_\pi^\lambda \\
&= \sum_{\tau \in C_{\sigma t}} \operatorname{sgn}(\sigma^{-1}\tau\sigma) \varphi_\sigma^\lambda \varphi_{\sigma^{-1}\tau\sigma}^\lambda \\
&= A_{\sigma t} \varphi_\sigma^\lambda
\end{aligned}
$$

Solution:

```
\documentclass{article}
\usepackage{amsmath}
\usepackage{amssymb}
\usepackage[a4paper,margin=1in]{geometry}
\begin{document}
\title{Mathematical Equations}
\author{}
\date{}
\maketitle
\begin{center}
% Left Side Equations (Quadratic Formula Steps)
```

```latex
    \begin{minipage}{0.45\textwidth}
    \begin{align*}
    x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\[6pt]
    &= \frac{-2 \pm \sqrt{2^2 - 4(1)(-8)}}{2 \cdot 1} \\[6pt]
    &= \frac{-2 \pm \sqrt{4 + 32}}{2}
    \end{align*}
    \end{minipage}
    \hfill
    % Right Side Equations (Summation Expressions)
    \begin{minipage}{0.45\textwidth}
    \begin{align*}
    \varphi_0^{\lambda} A_t
    &= \sum_{\pi \in C_t} \operatorname{sgn}(\pi)\,
    \varphi_0^{\lambda}\varphi_{\pi}^{\lambda} \\[8pt]
    &= \sum_{\tau \in C_{\sigma t}}
    \operatorname{sgn}(\sigma^{-1}\tau\sigma)\,
    \varphi_0^{\lambda}\varphi_{\sigma^{-1}\tau\sigma}^{\lambda} \\[8pt]
    &= A_{\sigma t}\varphi_0^{\lambda}
    \end{align*}
    \end{minipage}
    \end{center}
\end{document}
```

Output:

Mathematical Equations

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\[6pt] &= \frac{-2 \pm \sqrt{2^2 - 4(1)(-8)}}{2 \cdot 1} \\[6pt] &= \frac{-2 \pm \sqrt{4 + 32}}{2} \end{aligned}$$

$$\begin{aligned} \varphi_0^{\lambda} A_t &= \sum_{\pi \in C_t} \operatorname{sgn}(\pi)\, \varphi_0^{\lambda}\varphi_{\pi}^{\lambda} \\[8pt] &= \sum_{\tau \in C_{\sigma t}} \operatorname{sgn}(\sigma^{-1}\tau\sigma)\, \varphi_0^{\lambda}\varphi_{\sigma^{-1}\tau\sigma}^{\lambda} \\[8pt] &= A_{\sigma t}\varphi_0^{\lambda} \end{aligned}$$

8. Develop a LaTeX script to demonstrate the presentation of Numbered theorems, definitions, corollaries, and lemmas in the document.

Solution:

```
\documentclass{article}
\usepackage[a4paper,margin=1in]{geometry}
\usepackage{amsmath, amsthm}
% Theorem Environments
\newtheorem{theorem}{Theorem}
\newtheorem{lemma}{Lemma}
\newtheorem{corollary}{Corollary}
\theoremstyle{definition}
\newtheorem{definition}{Definition}
\begin{document}
\title{Example of Numbered Theorems, Definitions, Lemmas and Corollaries}
\author{}
\date{}
\maketitle
\section{Basic Number Theory}
\begin{definition}
An integer is called an even number if it is divisible by 2.
\end{definition}
\begin{theorem}
The sum of two even numbers is even.
\end{theorem}
 \begin{proof}
Let the two even numbers be $2a$ and $2b$, where $a$ and $b$ are integers.
Their sum is:
\[
2a + 2b = 2(a + b)
\]
Since $2(a+b)$ is divisible by 2, the sum is even.
\end{proof}
\begin{lemma}
If a number is divisible by 4, then it is even.
\end{lemma}
\begin{proof}
```

```
    Let the number be $4k$, where $k$ is an integer.
    Then,
    \[
    4k = 2(2k)
    \]
    Since it is divisible by 2, the number is even.
    \end{proof}
    \begin{corollary}
    The sum of two numbers divisible by 4 is also even.
    \end{corollary}
\end{document}
```

Output:

Example of Numbered Theorems, Definitions, Lemmas and Corollaries

1 Basic Number Theory

Definition 1. An integer is called an even number if it is divisible by 2.

Theorem 1. *The sum of two even numbers is even.*

Proof. Let the two even numbers be $2a$ and $2b$, where a and b are integers. Their sum is:

$$2a + 2b = 2(a + b)$$

Since $2(a + b)$ is divisible by 2, the sum is even. □

Lemma 1. *If a number is divisible by 4, then it is even.*

Proof. Let the number be $4k$, where k is an integer. Then,

$$4k = 2(2k)$$

Since it is divisible by 2, the number is even. □

Corollary 1. *The sum of two numbers divisible by 4 is also even.*

9. Develop a LaTeX script to create a document that consists of two paragraphs with a minimum of 10 citations in it and display the reference in the section

Solution:

```
\documentclass{article}
\usepackage[a4paper,margin=1in]{geometry}
\begin{document}
\section{Introduction}
Artificial Intelligence (AI) has become one of the most important
technologies in modern computing \cite{r1}. Machine learning
techniques are widely used in data analysis and prediction tasks
\cite{r2}. Deep learning models have significantly improved image
and speech recognition systems \cite{r3}. The development of neural
networks has transformed pattern recognition research \cite{r4}.
Many industries now depend on AI-driven automation systems
\cite{r5}.
In addition, AI applications are widely used in healthcare and
education \cite{r6}. Big data analytics plays a crucial role in
training intelligent systems \cite{r7}. Cloud computing supports
large-scale AI deployment \cite{r8}. Cybersecurity systems also use
machine learning algorithms for threat detection \cite{r9}.
Recent research continues to improve algorithm efficiency and
accuracy \cite{r10}.
\section{References}
\begin{thebibliography}{99}
\bibitem{r1} Russell, S. and Norvig, P., Artificial Intelligence: A Modern Approach, 2010.
\bibitem{r2} Mitchell, T., Machine Learning, 1997.
\bibitem{r3} Goodfellow, I., Deep Learning, 2016.
\bibitem{r4} Haykin, S., Neural Networks and Learning Machines, 2009.
\bibitem{r5} Ng, A., Machine Learning Yearning, 2018.
\bibitem{r6} Topol, E., Deep Medicine, 2019.
\bibitem{r7} Mayer-Schönberger, V., Big Data, 2013.
\bibitem{r8} Buyya, R., Cloud Computing, 2011.
\bibitem{r9} Stallings, W., Cryptography and Network Security, 2017.
\bibitem{r10} Murphy, K., Machine Learning: A Probabilistic Perspective, 2012.
\end{thebibliography}
```

```
\end{document}
```

Output:

1 Introduction

Artificial Intelligence (AI) has become one of the most important technologies in modern computing [1]. Machine learning techniques are widely used in data analysis and prediction tasks [2]. Deep learning models have significantly improved image and speech recognition systems [3]. The development of neural networks has transformed pattern recognition research [4]. Many industries now depend on AI-driven automation systems [5]. In addition, AI applications are widely used in healthcare and education [6]. Big data analytics plays a crucial role in training intelligent systems [7]. Cloud computing supports large-scale AI deployment [8]. Cybersecurity systems also use machine learning algorithms for threat detection [9]. Recent research continues to improve algorithm efficiency and accuracy [10].

2 References

References

[1] Russell, S. and Norvig, P., Artificial Intelligence: A Modern Approach, 2010.

[2] Mitchell, T., Machine Learning, 1997.

[3] Goodfellow, I., Deep Learning, 2016.

[4] Haykin, S., Neural Networks and Learning Machines, 2009.

[5] Ng, A., Machine Learning Yearning, 2018.

[6] Topol, E., Deep Medicine, 2019.

[7] Mayer-Schönberger, V., Big Data, 2013.

[8] Buyya, R., Cloud Computing, 2011.

[9] Stallings, W., Cryptography and Network Security, 2017.

[10] Murphy, K., Machine Learning: A Probabilistic Perspective, 2012.

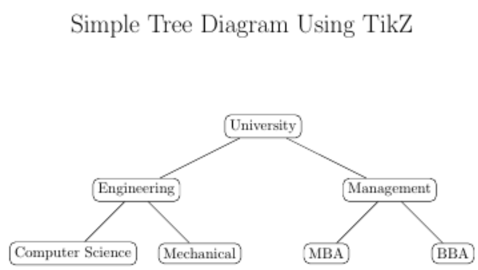
10. Develop a LaTeX script to design a simple tree diagram or hierarchical structure in the document with appropriate labels using the Tikz library.

Solution:

```
\documentclass{article}
\usepackage[a4paper,margin=1in]{geometry}
\usepackage{tikz}
\usetikzlibrary{trees}
\begin{document}
\title{Simple Tree Diagram Using TikZ}
\author{}
\date{}
\maketitle
\begin{center}
\begin{tikzpicture}[
level 1/.style={sibling distance=60mm},
level 2/.style={sibling distance=30mm},
every node/.style={draw, rectangle, rounded corners, align=center}
```

```
]
\node {University}
child { node {Engineering}
child { node {Computer Science} }
child { node {Mechanical} }
}
child { node {Management}
child { node {MBA} }
child { node {BBA} }
};
\end{tikzpicture}
\end{center}
\end{document}
```

Output:



11. Develop a LaTeX script to present an algorithm in the document using algorithm/algorithmic/algorithm2e library.

Solution:

```
\documentclass{article}
\usepackage[a4paper,margin=1in]{geometry}
\usepackage{algorithm}
\usepackage{algorithmic}
\begin{document}
\title{Linear Search Algorithm}
\author{}
\date{}
\maketitle
\section{Algorithm Example}
\begin{algorithm}
\caption{Linear Search}
```

```
\begin{algorithmic}[1]
\REQUIRE Array A of size n, Key element x
\ENSURE Position of x if found
\STATE Set position $\leftarrow -1$
\FOR{$i = 1$ to $n$}
\IF{$A[i] = x$}
\STATE position $\leftarrow i$
\STATE \textbf{break}
\ENDIF
\ENDFOR
\IF{position $\neq -1$}
\STATE Print ``Element found at position'', position
\ELSE
\STATE Print ``Element not found''
\ENDIF
\end{algorithmic}
\end{algorithm}
\end{document}
```

Output:

1 Algorithm Example

Algorithm 1 Linear Search

Require: Array A of size n, Key element x
Ensure: Position of x if found

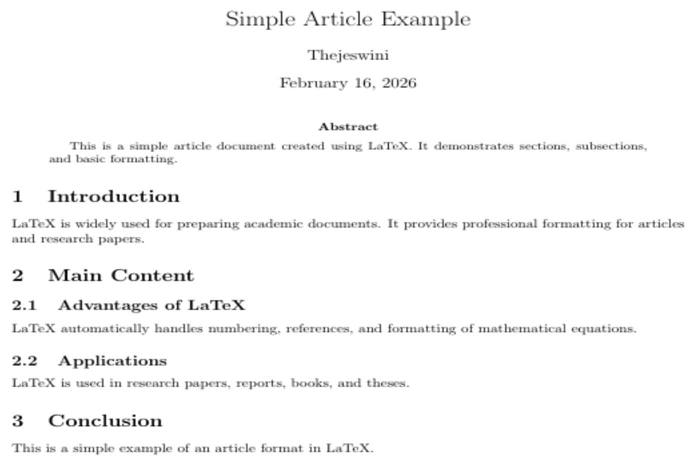
Set position $\leftarrow -1$
for $i = 1$ to n **do**
  if $A[i] = x$ **then**
    position $\leftarrow i$
    break
  end if
end for
if position $\neq -1$ **then**
  Print "Element found at position", position
else
  Print "Element not found"
end if

12. Develop a LaTeX script to create a simple report and article by using suitable commands and formats of user choice.

Solution:

```
    \documentclass{article}
    \usepackage[a4paper,margin=1in]{geometry}
    \title{Simple Article Example}
    \author{Student Name}
    \date{\today}
    \begin{document}
    \maketitle
    \begin{abstract}
    This is a simple article document created using LaTeX.
    It demonstrates sections, subsections, and basic formatting.
    \end{abstract}
    \section{Introduction}
    LaTeX is widely used for preparing academic documents.
    It provides professional formatting for articles and research papers.
    \section{Main Content}
    \subsection{Advantages of LaTeX}
    LaTeX automatically handles numbering, references,
    and formatting of mathematical equations.
    \subsection{Applications}
    LaTeX is used in research papers, reports, books, and theses.
    \section{Conclusion}
    This is a simple example of an article format in LaTeX.
\end{document}
```

Output:



Simple Article Example

Thejeswini

February 16, 2026

Abstract

This is a simple article document created using LaTeX. It demonstrates sections, subsections, and basic formatting.

1 Introduction

LaTeX is widely used for preparing academic documents. It provides professional formatting for articles and research papers.

2 Main Content

2.1 Advantages of LaTeX

LaTeX automatically handles numbering, references, and formatting of mathematical equations.

2.2 Applications

LaTeX is used in research papers, reports, books, and theses.

3 Conclusion

This is a simple example of an article format in LaTeX.

VIVA QUESTION

1. What is LaTeX?
Answer: LaTeX is a document preparation system used for typesetting documents with high-quality formatting, especially for scientific and technical documents.
2. How do you create a new section in LaTeX?
Answer: Use \section{Section Title} to create a new section in a LaTeX document.
3. What command is used to start a new paragraph in LaTeX?
Answer: Simply leave a blank line in the LaTeX source code to start a new paragraph.
4. How can you add bold text in LaTeX?
Answer: Use \textbf{} or \textit{} for bold or italic text, respectively.
5. What is the purpose of the \maketitle command?
Answer: \maketitle is used to generate a title based on the \title{}, \author{}, and \date{} commands.
6. How do you include figures in a LaTeX document?
Answer: Use the graphicx package and \includegraphics{} command to include images.
7. What environment is used for creating lists with bullet points in LaTeX?
Answer: The itemize environment is used for creating unordered lists.
8. How do you create a table in LaTeX?
Answer: Use the tabular environment to create tables with rows and columns.
9. What is a BibTeX file used for in LaTeX?
Answer: A BibTeX file (.bib) is used to store bibliographic data for citations and references.
10. How can you insert mathematical equations in a LaTeX document?
Answer: Use $...$ for inline equations and \[... \] for displayed equations.
11. What does the \cite{} command do in LaTeX?
Answer: The \cite{} command is used to insert citations from a BibTeX database.
12. What is the purpose of the \begin{document} and \end{document} commands?
Answer: They define the beginning and end of the document content in a LaTeX file.
13. How do you create subsections in LaTeX?
Answer: Use \subsection{Subsection Title} to create subsections within a section.
14. What is the purpose of the enumerate environment in LaTeX?
Answer: The enumerate environment is used for creating numbered lists.
15. How can you include comments in a LaTeX document?
Answer: Use the % symbol to insert comments, which are ignored by the compiler.
16. What does the article document class represent in LaTeX?
Answer: The article document class is used for shorter documents such as journal articles.
17. How do you add a header and footer to a LaTeX document?
Answer: Use the \pagestyle{} command to specify headers and footers for a document.
18. How can you change the font size in LaTeX?
Answer: Use commands like \small, \large, or \fontsize{size}{skip} to change font sizes.
19. What package is commonly used for including code snippets in LaTeX documents?
Answer: The listings package is often used for typesetting code.
20. How do you create a title page in LaTeX?
Answer: Use \title{}, \author{}, \date{}, and \maketitle to format a title page.
21. What is the purpose of the \tableofcontents command in LaTeX?
Answer: \tableofcontents generates a table of contents based on section headings.
22. How do you add a line break in LaTeX?
Answer: Use \\ or \newline to insert a line break.

23. What is the purpose of the amsmath package in LaTeX?
Answer: The amsmath package provides enhanced mathematical formatting and commands.
24. How can you create footnotes in LaTeX?
Answer: Use the \footnote{} command to insert footnotes in a document.
25. What does the geometry package do in LaTeX?
Answer: The geometry package allows customization of page layout and margins.

26. How do you create cross-references in LaTeX?
Answer: Use \label{} to mark a reference point and \ref{} to refer to it elsewhere in the document.
27. What is the purpose of the \include{} and \input{} commands in LaTeX?
Answer: They are used to include external files into the main LaTeX document.
28. How can you create emphasized (italicized) text in LaTeX?
Answer: Use \emph{} to emphasize text by making it italicized.
29. What is the role of the .aux file in LaTeX document compilation?
Answer: The .aux file contains auxiliary information used for cross-referencing and citation management.
30. How do you generate a bibliography in LaTeX?
Answer: Use \bibliography{} to specify a .bib file and \bibliographystyle{} to format the bibliography style.